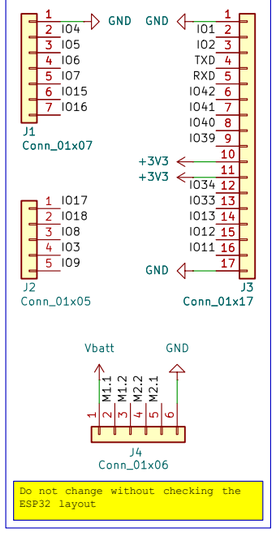
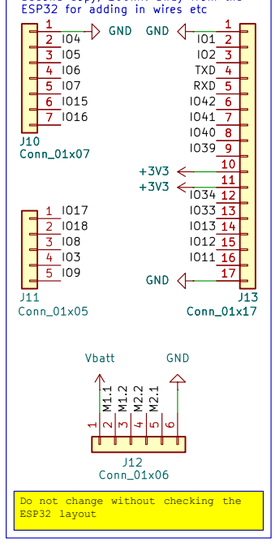
Header IOs for the ESP32



Do not change without checking the ESP32 layout

Second copy, 100mil away from the ESP32 for adding in wires etc

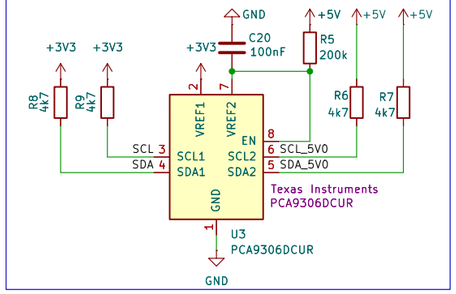


Do not change without checking the ESP32 layout

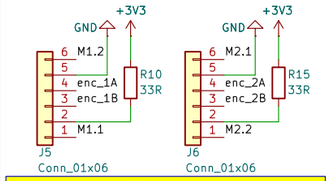
IO Mapping

IO1	Servo_1
IO2	Servo_2
IO3	enc_1B
IO4	LED_R
IO5	LED_G
IO6	LED_B
IO7	LED_Y
IO8	SW_1
IO9	enc_1A
IO11	enc_2B
IO12	enc_2A
IO13	SW_2
IO15	LF_8
IO16	LF_7
IO17	LF_6
IO18	LF_5
IO33	SCL
IO34	SDA
IO39	LF_4
IO40	LF_3
IO41	LF_2
IO42	LF_1

I2C pull ups and level shift to 5V0

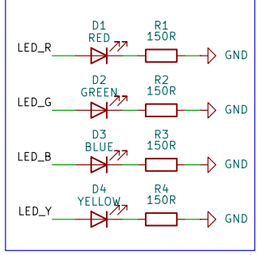


Motor with encoder connecion via JST-XH

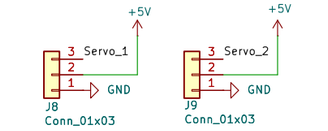


Motors use JST ZH 6 way connectors.
Note different wiring. Can reverse in SW

Four LEDs (R, G, B, Y)

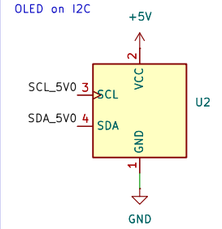


Two Servo Connectors



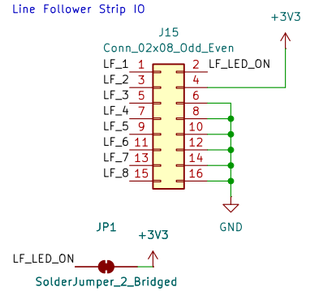
Most servos can take 3V3 PWM

OLED on I2C



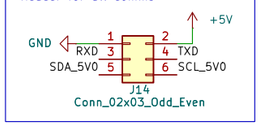
ER_OLEDM0.91_1x-I2C

Line Follower Strip IO

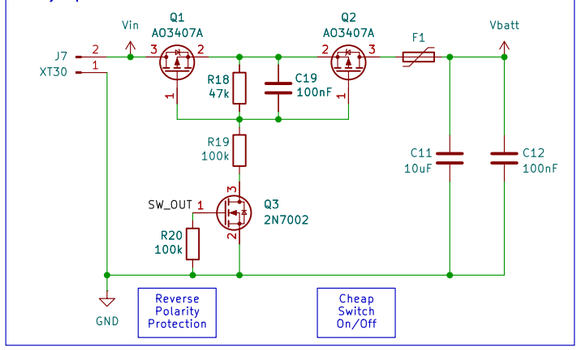


LEDs on permanantly by default. No spare IO

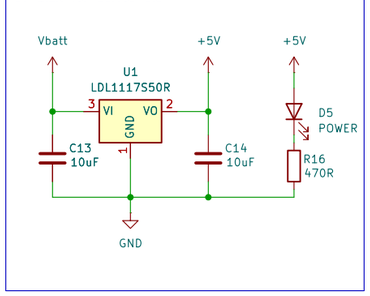
Header for all comms



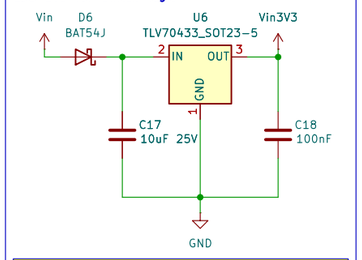
Battery Input and Protection



5V 1.2A for Servos etc

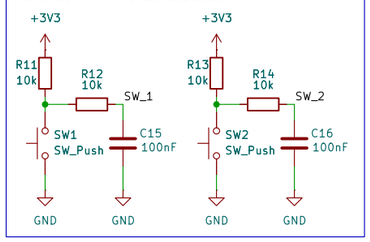


LDO 3V3 for Switch logic

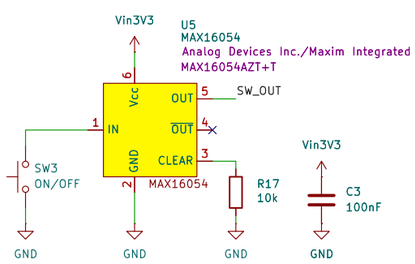


Always needs to be on so straight from Input

Switches with hardware debounce



One switch soft latching



QUT U4
ESP32 MotorDriver BOB_v02

MountingHole
H1
H3
H2
H4

Break Out Board
- Headers for Microcontroller board, with a socket for each pin to be broken out if needed
- XT30 connector, allowing a 2S battery, therefore needing a 5V0 regulator and maybe 3V3
- 2 JST ZH headers for the motor with encoder connection
- 2 Servo connector headers
- I2C header
- SPI header
- UART header
- Header for the LineFollower Sparkfun strip
- Header for the I2C OLED, or have OLED on board
- LEDS x 4 (one each of RED, GREEN, BLUE and YELLOW)
- Two larger buttons for digital input to the micro
- PCB should be RED

Sheet: /
File: ESP32_MotorDriver_BOB.kicad_sch

Title:

Size: A3	Date:	**Rev: v02**
KiCad E.D.A. 10.0.1		Id: 1/1